

第二章 平面解析几何

2.6.2 双曲线的几何性质

【纯答案册】题面不重印 印面号与正册同号同序

1. [答案] (1) 实轴长 $8\sqrt{2}$, 虚轴长 4, 顶点 $(-4\sqrt{2}, 0)$ 和 $(4\sqrt{2}, 0)$; (2) 实轴长 6, 虚轴长 18, 顶点 $(-3, 0)$ 和 $(3, 0)$; (3) 实轴长 4, 虚轴长 4, 顶点 $(0, -2)$ 和 $(0, 2)$; (4) 实轴长 10, 虚轴长 14, 顶点 $(0, -5)$ 和 $(0, 5)$ 。

[详解] (1) 化 $x^2/32 - y^2/4 = 1$, 焦点在 x 轴, $a^2 = 32$, $b^2 = 4$, $a = 4\sqrt{2}$, $b = 2$: 实轴长 $2a = 8\sqrt{2}$, 虚轴长 $2b = 4$, 顶点 $(-4\sqrt{2}, 0)$ 和 $(4\sqrt{2}, 0)$ 。

(2) 化 $x^2/9 - y^2/81 = 1$, 焦点在 x 轴, $a = 3$, $b = 9$: 实轴长 6, 虚轴长 18, 顶点 $(-3, 0)$ 和 $(3, 0)$ 。

(3) 化 $y^2/4 - x^2/4 = 1$, 焦点在 y 轴, $a = 2$, $b = 2$: 实轴长 4, 虚轴长 4, 顶点 $(0, -2)$ 和 $(0, 2)$ 。

(4) 化 $y^2/25 - x^2/49 = 1$, 焦点在 y 轴, $a = 5$, $b = 7$: 实轴长 10, 虚轴长 14, 顶点 $(0, -5)$ 和 $(0, 5)$ 。

2. [答案] 实轴长 18; 虚轴长 6; 焦点 $(0, \pm 3\sqrt{10})$; 离心率 $e = \sqrt{10}/3$; 渐近线方程 $y = \pm 3x$ 。

[详解] 化 $-9x^2 + y^2 = 81$ 为 $y^2/81 - x^2/9 = 1$, 焦点在 y 轴, $a^2 = 81$, $b^2 = 9$, $a = 9$, $b = 3$, $c^2 = a^2 + b^2 = 90$, $c = \sqrt{90} = 3\sqrt{10}$ 。

于是: 实轴长 $2a = 18$; 虚轴长 $2b = 6$; 焦点 $(0, \pm 3\sqrt{10})$; 离心率 $e = c/a = 3\sqrt{10}/9 = \sqrt{10}/3$; 渐近线 $y = \pm(a/b)x = \pm 3x$ 。

验算: $e = \sqrt{1 + (a/b)^2} = \sqrt{10}/3$ 合; $2c = 6\sqrt{10} \approx 18.97 > 2a = 18$ 合; 点 $(0, 9)$ 代入方程 $81/81 = 1$ 合。

3. [答案] D

[详解] 实轴长 8 得 $a = 4$; 焦点在 y 轴, 渐近线 $y = \pm(a/b)x = \pm(4/b)x = (4/3)x$, 得 $b = 3$. 故 C: $y^2/16 - x^2/9 = 1$, 选 D。

4. [答案] (1) 实轴长 $8/3$, 虚轴长 4, 焦点 $(\pm 2\sqrt{13}/3, 0)$, 顶点 $(\pm 4/3, 0)$, 渐近线 $y = \pm(3/2)x$; (2) ($m > 0$) 实轴长 4, 虚轴长 $2\sqrt{m}$, 焦点 $(\pm\sqrt{m+4}, 0)$, 顶点 $(\pm 2, 0)$, 渐近线 $y = \pm(\sqrt{m}/2)x$ 。

[详解] (1) $a^2 = 16/9$, $b^2 = 4$, $a = 4/3$, $b = 2$; $c^2 = 16/9 + 4 = 52/9$, $c = \sqrt{52}/3 = 2\sqrt{13}/3$; 实轴长 $8/3$, 虚轴长 4, 焦点 $(\pm 2\sqrt{13}/3, 0)$, 顶点 $(\pm 4/3, 0)$, 渐近线 $y = \pm(b/a)x = \pm(3/2)x$ 。

(2) 双曲线须 $m > 0$: $a = 2$, $b = \sqrt{m}$, $c^2 = 4 + m$, $c = \sqrt{m+4}$: 实轴长 4, 虚轴长 $2\sqrt{m}$, 焦点 $(\pm\sqrt{m+4}, 0)$, 顶点 $(\pm 2, 0)$, 渐近线 $y = \pm(\sqrt{m}/2)x$. ($m < 0$ 时方程表椭圆, 与题设矛盾, 仅 $m > 0$ 一解.)

5. [答案] 实轴长 2; 虚轴长 $4\sqrt{6}$; 焦点 $(\pm 5, 0)$; 渐近线方程 $y = \pm 2\sqrt{6}x$ 。

[详解] $a^2 = 1$, $b^2 = 24$, $a = 1$, $b = 2\sqrt{6}$, $c^2 = 1 + 24 = 25$, $c = 5$: 实轴长 2, 虚轴长 $4\sqrt{6}$, 焦点 $(\pm 5, 0)$, 渐近线 $y = \pm(b/a)x = \pm 2\sqrt{6}x$ 。

6. [答案] $x^2/16 - y^2/9 = 1$; $e = 5/4$ 。

[详解] 焦点 $(5, 0)$ 在 x 轴: $b/a = 3/4$, $c = 5$, $a^2 + b^2 = 25$, $b^2 = (9/16)a^2$ 代入得 $a^2(1 + 9/16) = 25$, $a^2 = 16$, $b^2 = 9$. 标准方程 $x^2/16 - y^2/9 = 1$, $e = c/a = 5/4$ 。

7. [答案] 它们的渐近线相同, 均为 $y = \pm(b/a)x$ 。

[详解] $\lambda > 0$ 时方程即 $x^2/(a^2\lambda) - y^2/(b^2\lambda) = 1$, 焦点在 x 轴, 渐近线由 $x^2/(a^2\lambda) - y^2/(b^2\lambda) = 0$ 得 $y = \pm(b/a)x$; $\lambda < 0$ 时方程即 $y^2/(-b^2\lambda) - x^2/(-a^2\lambda) = 1$, 焦点在 y 轴, 渐近线由同式得 $y = \pm(b/a)x$. 故所

有曲线的渐近线相同, 均为 $y = \pm(b/a)x$ (渐近线只取决于 $a:b$, 与 λ 无关).

8. [答案] $x^2/9 - y^2/27 = 1$ (焦点在 x 轴; 焦点在 y 轴时为 $y^2/9 - x^2/27 = 1$).

[详解] 顶点距 6, 即 $2a = 6, a = 3$; 焦点段 $[-c, c]$ 上的分点 $-a, 0, a$ 将其四等分, 等分距为 a , 故 $a = c/2, c = 6, b^2 = c^2 - a^2 = 36 - 9 = 27$. 焦点在 x 轴: $x^2/9 - y^2/27 = 1$; 题面未指明焦点位置, 焦点在 y 轴时 $y^2/9 - x^2/27 = 1$ 同法可得.

9. [答案] B

[详解] 由 $x^2 - y^2/4 = 1$ 知焦点在 x 轴, $a = 1, b = 2$, 渐近线 $y = \pm(b/a)x = \pm 2x$, 故选 B.

10. [答案] 3

[详解] $c^2 = 16 + 9 = 25, c = 5$, 焦点 $(\pm 5, 0)$; 渐近线方程 $3x \pm 4y = 0$. 取 $F(5, 0)$ 与直线 $3x - 4y = 0$: $d = |3 \times 5 - 4 \times 0| / \sqrt{3^2 + 4^2} = 15/5 = 3$. 由对称性, 任一焦点到任一条渐近线的距离均为 3 (恰为半虚轴长 $b = 3$).

11. [答案] B

[详解] $F(-c, 0), A(a, 0)$, 取 $B(0, b)$, 直线 $AB: bx + ay - ab = 0$. F 到 AB 的距离 $d = |-bc - ab| / \sqrt{a^2 + b^2} = b(c + a)/c$ (分子提出 b , 分母用 $c^2 = a^2 + b^2$).

由 $d = (5/4)b$ 得 $c + a = (5/4)c$, 即 $4a = c; b^2 = c^2 - a^2 = 15a^2, b/a = \sqrt{15}$, 渐近线 $y = \pm\sqrt{15}x$, 故选 B. 验算: $c = 4a$ 时 $d = b \cdot 5a / (4a) = (5/4)b$ 合.

12. [答案] B

[详解] 共渐近线设 $x^2 - 2y^2 = m (m \neq 0)$; 焦点 $(0, -6)$ 在 y 轴上, 故 $m < 0$. 标准形 $y^2/(-m/2) - x^2/(-m) = 1$: $a^2 = -m/2, b^2 = -m, c^2 = a^2 + b^2 = -3m/2 = 36$, 得

$m = -24$, 方程 $y^2/12 - x^2/24 = 1$, 选 B. 验算: $c^2 = 12 + 24 = 36$ 合, 焦点含 $(0, -6)$ 合.

13. [答案] $x^2/(32/5) - y^2/(18/5) = 1$.

[详解] 椭圆 $a^2 = 10, b^2 = 5, c^2 = 5$, 长轴端点 $(\pm\sqrt{10}, 0)$, 即双曲线焦点在 x 轴, $c = \sqrt{10}$; 渐近线 $y = \pm(3/4)x, b/a = 3/4. a^2 + b^2 = 10, a^2(1 + 9/16) = 10$, 得 $a^2 = 32/5, b^2 = 10 - 32/5 = 18/5$, 标准方程 $x^2/(32/5) - y^2/(18/5) = 1$.

14. [答案] $x^2/(64/13) - y^2/(144/13) = 1$.

[详解] 焦点在 x 轴, $c = 4$; 渐近线 $y = (3/2)x, b/a = 3/2, a^2 + b^2 = 16; b^2 = (9/4)a^2$ 代入得 $a^2(13/4) = 16, a^2 = 64/13, b^2 = 16 - 64/13 = 144/13$, 标准方程 $x^2/(64/13) - y^2/(144/13) = 1$.

15. [答案] $x^2/12 - y^2/8 = 1$.

[详解] 原双曲线 $c^2 = 16 + 4 = 20$, 有公共焦点, 故所求双曲线 $c^2 = 20$, 焦点在 x 轴; 半实轴长 $a = 2\sqrt{3}, a^2 = 12, b^2 = c^2 - a^2 = 20 - 12 = 8$, 标准方程 $x^2/12 - y^2/8 = 1$.

16. [答案] 2

[详解] 焦点在 x 轴, 渐近线 $y = \pm(b/a)x$, 故 $b/a = \sqrt{3}. e = c/a = \sqrt{1 + (b/a)^2} = \sqrt{1 + 3} = 2$.

17. [答案] $5/4$ (焦点在 x 轴) 或 $5/3$ (焦点在 y 轴)

[详解] 焦点在 x 轴时, 渐近线 $y = \pm(b/a)x, b/a = 3/4, e = \sqrt{1 + (b/a)^2} = \sqrt{1 + 9/16} = \sqrt{25/16} = 5/4$; 焦点在 y 轴时, 渐近线 $y = \pm(a/b)x, a/b = 3/4, b/a = 4/3, e = \sqrt{1 + 16/9} = \sqrt{25/9} = 5/3$. 题面未限焦点位置, 两解并立.

18.[答案] $x^2/3 - y^2 = 1$

[详解] 由渐近线得 $b/a = \sqrt{3}/3$, 即 $a = \sqrt{3}b$. 直线 l 过 $P(0, b)$ 、 $Q(a, 0)$, 方程 $bx + ay - ab = 0$, 原点到 l 的距离 $ab/\sqrt{a^2 + b^2} = \sqrt{3}/2$, 两边平方得 $4a^2b^2 = 3(a^2 + b^2)$. 联立 $a = \sqrt{3}b$: $4 \times 3b^4 = 3(3b^2 + b^2)$, 即 $12b^4 = 12b^2$, $b^2 = 1 (b > 0)$, $a^2 = 3$. 故双曲线方程为 $x^2/3 - y^2 = 1$. 验算: $a = \sqrt{3}$ 、 $b = 1$ 时 $ab/\sqrt{a^2 + b^2} = \sqrt{3}/2$ 合, 渐近线 $y = \pm x/\sqrt{3}$ 合.

19.[答案] (1) $x^2/9 - y^2/25 = 1$ 与 $y^2/25 - x^2/9 = 1$ (答案不唯一, 凡渐近线为 $5x \pm 3y = 0$ 的标准方程均可); (2) $y^2/(56/9) - x^2/(56/25) = 1$ (即 $9y^2 - 25x^2 = 56$).

[详解] (1) 两渐近线即 $y = \pm(5/3)x$. 焦点在 x 轴: $b/a = 5/3$, 取 $a = 3$ 、 $b = 5$ 得 $x^2/9 - y^2/25 = 1$; 焦点在 y 轴: $a/b = 5/3$, 取 $a = 5$ 、 $b = 3$ 得 $y^2/25 - x^2/9 = 1$.

(2) 以 l_1 、 l_2 为渐近线的双曲线可设为 $x^2/9 - y^2/25 = \lambda$ ($\lambda \neq 0$). 代入 $M(1, 3)$: $\lambda = 1/9 - 9/25 = (25 - 81)/225 = -56/225 < 0$, 故焦点在 y 轴, 方程即 $y^2/25 - x^2/9 = 56/225$, 化标准形 $y^2/(56/9) - x^2/(56/25) = 1$. 验算: $9y^2 - 25x^2 = 56$ 代入 $M(1, 3)$: $81 - 25 = 56$ 合. 提醒: 渐近线同为 $5x \pm 3y = 0$ 的双曲线焦点可在 x 轴或 y 轴, 由所过点的坐标定号 (本题 $1/9 - 9/25 < 0$ 定 y 轴).

20.[答案] 曲线是双曲线 (两支): 以 AB 所在直线为 x 轴、 AB 中点为原点, 其方程为 $x^2/9 - y^2/16 = 1$.

[详解] 曲线上点 P 满足 $||PA| - |PB|| = 6 < |AB| = 10$, 由定义轨迹为双曲线, $2a = 6$ 、 $2c = 10$, $a = 3$ 、 $c = 5$, $b^2 = 25 - 9 = 16$. 以 AB 中点为原点、 AB 所在直线为 x 轴建系, 方程为 $x^2/9 - y^2/16 = 1$. 作图观察: 交点连成两支光滑曲线, 与双曲线吻合.

21.[答案] $x^2/4 - y^2/5 = 1$.

[详解] (直接法) 设 $M(x, y)$: $\sqrt{(x-3)^2 + y^2} = (3/2)|x - 4/3|$. 两边平方: $x^2 - 6x + 9 + y^2 = (9/4)(x^2 - (8/3)x + 16/9) = (9/4)x^2 - 6x + 4$.

$-6x$ 抵消: $y^2 = (9/4)x^2 + 4 - x^2 - 9 = (5/4)x^2 - 5$, 即 $(5/4)x^2 - y^2 = 5$, 标准形 $x^2/4 - y^2/5 = 1$. 验算: $a = 2$ 、 $c = 3$ 与准线位置 $x = a^2/c = 4/3$ 、比值 $c/a = 3/2$ 相容合.

22.[答案] 比值等于 c/a (离心率 e).

[详解] (直接法) $P(x, y)$ 在双曲线上: $y^2 = b^2(x^2/a^2 - 1)$. $d_1 = |PF| = \sqrt{(x+c)^2 + y^2}$, 代入并用 $b^2 = c^2 - a^2$ 化简: $x^2 + 2cx + c^2 + (c^2 - a^2)x^2/a^2 - (c^2 - a^2) = (c^2/a^2)x^2 + 2cx + a^2 = [(c/a)x + a]^2$.
 $d_2 = |x + a^2/c|$ (P 到 l 的距离). 故 $d_1/d_2 = (c/a)|x + a^2/c|/|x + a^2/c| = c/a$ (即离心率 e).

23.[答案] $x^2/a^2 - y^2/b^2 = 1$, 其中 $b^2 = c^2 - a^2$.

[详解] (直接法) 设 $M(x, y)$: $\sqrt{(x+c)^2 + y^2} = (c/a)|x + a^2/c|$. 平方: $x^2 + 2cx + c^2 + y^2 = (c^2/a^2)x^2 + 2cx + a^2$, $2cx$ 抵消: $y^2 = (c^2/a^2 - 1)x^2 - (c^2 - a^2) = (b^2/a^2)x^2 - b^2$ ($b^2 = c^2 - a^2$), 即 $x^2/a^2 - y^2/b^2 = 1$. 解释: 2.6.2 例 3 所得轨迹方程正是 $x^2/a^2 - y^2/b^2 = 1$ ($b^2 = c^2 - a^2$), 本题表明该双曲线可由“到定点与到定直线距离之比为常数 c/a ”直接刻画.

24.[答案] A

[详解] 设 $M(x_0, y_0)$, 则 $N(-x_0, -y_0)$, $P(x, y)$ ($x \neq \pm x_0$).
 $k_1 k_2 = \frac{y - y_0}{x - x_0} \cdot \frac{y + y_0}{x + x_0} = \frac{y^2 - y_0^2}{x^2 - x_0^2}$ (坐标点差法, 不引名目). M 、 P 均在双曲线上: $y^2 = b^2(x^2/a^2 - 1)$, $y_0^2 = b^2(x_0^2/a^2 - 1)$, 相减得 $y^2 - y_0^2 = (b^2/a^2)(x^2 - x_0^2)$, 故 $k_1 k_2 = b^2/a^2$ (定值).
于是 $|k_1| + |k_2| \geq 2\sqrt{|k_1 k_2|} = 2b/a$ (当 $|k_1| = |k_2| = b/a$ 时取等, P 可取到), 由最小值 2 得 $b/a = 1$, $e = \sqrt{1 + (b/a)^2} = \sqrt{2}$, 故选 A.

25.[答案] 1

[详解] $c^2 = 16 + 9 = 25$, $c = 5$, 右焦点 $F(5, 0)$. 设 $P(x, y)$, $|PF|^2 = (x-5)^2 + y^2$. 由 $y^2 = 9(x^2/16 - 1)$ ($|x| \geq 4$):
 $|PF|^2 = x^2 - 10x + 25 + (9/16)x^2 - 9 = (25/16)x^2 - 10x + 16$.

对称轴 $x = 10/(2 \times 25/16) = 16/5 < 4$, 故在 $|x| \geq 4$ 上于 $x = 4$ 取最小: $|PF|^2 = (25/16) \times 16 - 40 + 16 = 1$, $|PF|_{\min} = 1$ (即右顶点处取得, 同支焦半径最小值 $c - a = 5 - 4 = 1$ 合).

26.[答案] 证明见解析.

[详解] 证明: 不妨先设焦点在 x 轴: 双曲线 $x^2/a^2 - y^2/b^2 = 1$ ($a > 0, b > 0$), 焦点 $F(c, 0)$, 渐近线 $y = \pm(b/a)x$, 即 $bx \mp ay = 0$. 焦点到渐近线的距离 $d = |bc - a \cdot 0|/\sqrt{a^2 + b^2} = bc/c = b$ (半虚轴长). 焦点在 y 轴时 ($y^2/a^2 - x^2/b^2 = 1$, 焦点 $F(0, c)$, 渐近线 $y = \pm(a/b)x$ 即 $ay \mp bx = 0$): $d = |a \cdot 0 - bc|/\sqrt{a^2 + b^2} = bc/c = b$, 等式仍成立. 证毕.

27.[答案] 2

[详解] 设 $P(x, y)$ ($|x| \geq 1$): $|PA|^2 = (x-3)^2 + y^2 = (x-3)^2 + 3(x^2-1) = 4x^2 - 6x + 6$. 对称轴 $x = 3/4 < 1$, 故 $x = 1$ (右顶点) 时取最小: $4 - 6 + 6 = 4$, $|PA|_{\min} = 2$.

28.[答案] 16

[详解] $a = 3, b = 4, c = 5$. 设两焦半径 $r_1 = |MF_1|, r_2 = |MF_2|$ ($r_1 > r_2$). $MF_1 \perp MF_2: r_1^2 + r_2^2 = |F_1F_2|^2 = 100$; 定义: $r_1 - r_2 = 2a = 6$. $(r_1 - r_2)^2 = 100 - 2r_1r_2 = 36$, 得 $r_1r_2 = 32$. $S = (1/2)r_1r_2 \sin 90^\circ = 16$.

29.[答案] (1) $ax^2 - y^2 = 36a$ ($y \neq 0$); (2) $a > 0$ 时为焦点在 x 轴的双曲线 (除去顶点); $a < 0$ 时为椭圆; 其中 $a = -1$ 为圆 $x^2 + y^2 = 36$ ($y \neq 0$), $-1 < a < 0$ 为焦点在 x 轴的椭圆, $a < -1$ 为焦点在 y 轴的椭圆 (均除去长轴端点——方程中 $y \neq 0$).

[详解] (1) 过 $(6, 0)$ 斜率 k_1 , 过 $(-6, 0)$ 斜率 $k_2, k_1k_2 = a$. 设 $M(x, y): k_1 = y/(x-6), k_2 = y/(x+6)$ ($x \neq \pm 6$, 且 M 不与 $(\pm 6, 0)$ 重合故 $y \neq 0$). $[y/(x-6)] \cdot [y/(x+6)] = a$, 得 $y^2 = a(x^2 - 36)$, 即 $ax^2 - y^2 = 36a$ ($y \neq 0$).

(2) $a > 0: x^2/36 - y^2/(36a) = 1$, 焦点在 x 轴的双曲线 (除去顶点); $a < 0$: 记 $a = -|a|$, $x^2/36 + y^2/(36|a|) = 1$, 椭圆; 其中 $a = -1: x^2 + y^2 = 36$, 圆; $-1 < a < 0: 36 > 36|a|$, 焦点在 x 轴的椭圆; $a < -1: 36 < 36|a|$, 焦点在 y 轴的椭圆 (均除去 x 轴上的点, 即 $y \neq 0$).

30.[答案] 证明见解析.

[详解] 证明: 设 $C: x^2/m^2 - y^2/m^2 = 1$ ($m > 0$), 则 $F_1(-\sqrt{2}m, 0), F_2(\sqrt{2}m, 0)$. 设 $P(x, y)$, 满足 $x^2 - y^2 = m^2$. $|PO|^2 = x^2 + y^2$. $|PF_1| \cdot |PF_2| = \sqrt{(x + \sqrt{2}m)^2 + y^2} \cdot \sqrt{(x - \sqrt{2}m)^2 + y^2}$, 根号内合并为 $\sqrt{(x^2 + 2m^2 + y^2)^2 - 8m^2x^2}$. 用 $y^2 = x^2 - m^2$ 化简: $x^2 + 2m^2 + y^2 = 2x^2 + m^2, (2x^2 + m^2)^2 - 8m^2x^2 = 4x^4 - 4m^2x^2 + m^4 = (2x^2 - m^2)^2$. 故 $|PF_1| \cdot |PF_2| = |2x^2 - m^2| = 2x^2 - m^2$ ($|x| \geq m$ 时为正) $= x^2 + y^2 = |PO|^2$. 证毕. 验算: $m = 1, P(\sqrt{2}, 1): |PO|^2 = 3, |PF_1| = 3, |PF_2| = 1$, 积为 3 合.

31.[答案] B

[详解] PQ 为通径, $P(c, \pm b^2/a), |PQ| = 2b^2/a$. $\triangle PF_1Q$ 中 $\angle PF_1Q = 90^\circ, F_2$ 为斜边 PQ 的中点, 由直角三角形斜边中线定理 $|F_1F_2| = |PQ|/2$, 即 $2c = b^2/a, 2ac = b^2 = c^2 - a^2$. 两边除以 $a^2: e^2 - 2e - 1 = 0, e > 1, e = 1 + \sqrt{2}$, 故选 B.

32.[答案] 2

[详解] A 在线段 F_1B 上且 $|F_1A| = |AB|$, 故 A 为 F_1B 的中点; O 为 F_1F_2 的中点, 故 OA 是 $\triangle F_1F_2B$ 的中位线: $OA \parallel F_2B, |F_2B| = 2|OA|$. 由 $\overrightarrow{F_1B} \cdot \overrightarrow{F_2B} = 0$ 得 $F_1B \perp F_2B$, 结合 $OA \parallel F_2B$ 得 $OA \perp F_1B$. 于是 $\triangle F_1OB$ 中, OA 既是边 F_1B 上的中线又是高, 故 $\triangle F_1OB$ 等腰: $|OB| = |OF_1| = c$, 且 $\angle AOF_1 = \angle AOB$. 又 A, B 分别在两条渐近线上, 两渐近线关于 x 轴对称, 故 $\angle BOF_2 = \angle AOF_1$; 而 $\angle AOF_1 + \angle AOB + \angle BOF_2 =$

180° , 三者在上述等量关系下相等, 各为 60° . 故 $b/a = \tan 60^\circ = \sqrt{3}$, $e = \sqrt{1+(b/a)^2} = \sqrt{1+3} = 2$.

33.[答案] C

[详解] 六个顶点内接于以 F_1F_2 为直径的圆, 由直径所对圆周角 $\angle F_1PF_2 = 90^\circ$; 正六边形各边对圆心角 60° , 得 $\angle PF_2F_1 = 30^\circ$. 解三角形: $|PF_2| = 2c \cos 30^\circ = \sqrt{3}c$, $|PF_1| = 2c \sin 30^\circ = c$. P 在双曲线上, $|PF_2| - |PF_1| = 2a$, 即 $\sqrt{3}c - c = 2a$, $e = c/a = 2/(\sqrt{3}-1) = \sqrt{3}+1$, 故选 C.

34.[答案] D

[详解] 圆 $x^2 + y^2 = a^2 + b^2$ 的半径为 c , 其直径恰为 F_1F_2 , 故 P 在圆上, $\angle F_1PF_2 = 90^\circ$ (直径所对圆周角). 设 $|PF_1| = n$, 由定义 $|PF_2| = 2a + n$. Rt $\triangle PF_1F_2$ 中: $n^2 + (2a+n)^2 = 4c^2$, 展开得 $n^2 = 2b^2 - 2an$.

由 $\overrightarrow{F_1P} = (1/2)\overrightarrow{F_2Q}$ (同向向量) 得 $|F_2Q| = 2n$, 且 $F_1P \parallel F_2Q$, 故 $\angle QF_2F_1 = 180^\circ - \angle PF_1F_2$, $\cos \angle QF_2F_1 = -n/(2c)$. 由定义 $|F_1Q| = 2a + 2n$, $\triangle F_1F_2Q$ 中余弦定理: $(2a+2n)^2 = (2n)^2 + (2c)^2 + 4n^2$, 化简得 $2an = b^2 + n^2$, 即 $b^2 = 2an - n^2$.

与 $n^2 = 2b^2 - 2an$ 联立: $3n^2 = 2an$, $a = (3/2)n$; $b^2 = 2n^2$, $c^2 = (17/4)n^2$, $e = c/a = \sqrt{17}/3$, 故选 D.

35.[答案] $\sqrt{3}$

[详解] $O(0,0)$, $A(-a,0)$, $B(a,0)$, 圆 P 的圆心为 $P(0,a)$, 半径 $\sqrt{2}a$. $|OA| = |OB| = |OP| = a$, 故 O 、 A 、 P 、 B 四点共圆 (圆心 O 、半径 a), 其中 AB 为该圆的直径, P 在圆上, 由直径所对圆周角得 $\angle APB = 90^\circ$. M 在圆上且在第一象限, 位于弦 AB 所对优弧, 故 $\angle AMB = (1/2)\angle APB = 45^\circ$.

夹角公式: $\tan \angle AMB = (n-m)/(1+nm) = 1$, $n-m = 3$, 得 $nm = 2$. 设 $M(x,y)$: $m = y/(x+a)$, $n = y/(x-a)$, $nm = y^2/(x^2 - a^2)$; 由 $y^2 = b^2(x^2/a^2 - 1)$ 得 $nm = b^2/a^2 = 2$. $e^2 = 1 + b^2/a^2 = 3$, $e = \sqrt{3}$.

36.[答案] $2\sqrt{3}/3$

[详解] 设 $A(a,0)$, $B(-a,0)$, $P(c,y)$ ($l: x = c$). $k_{PA} = y/(c-a)$, $k_{PB} = y/(c+a)$. 由夹角公式 $\tan \angle APB = |[k_{PA} - k_{PB}]/[1 + k_{PA}k_{PB}]| = \sqrt{3}$: $|(2ay)/(c^2 - a^2)| / |1 + y^2/(c^2 - a^2)| = \sqrt{3}$, 整理为关于 y 的方程 $\sqrt{3}y^2 - 2ay + \sqrt{3}(c^2 - a^2) = 0$ 有实根. 判别式 $\Delta = 4a^2 - 12(c^2 - a^2) = 4(4a^2 - 3c^2) \geq 0$, 得 $e^2 \leq 4/3$, $e \leq 2\sqrt{3}/3$. 离心率最大值 $2\sqrt{3}/3$. 验算: $e = 2\sqrt{3}/3$ 时 $\Delta = 0$, P 取对称轴位置 $y = a/\sqrt{3}$ 合.

37.[答案] 18

[详解] PQ 与 F_1F_2 互相平分 (P 、 Q 关于 O 对称, F_1 、 F_2 关于 O 对称) 且 $|PQ| = |F_1F_2|$, 对角线相等又互相平分, 故四边形 PF_1QF_2 为矩形, 其面积 $= |PF_1| \cdot |PF_2|$. 矩形中 $|PF_1|^2 + |PF_2|^2 = |F_1F_2|^2 = 4c^2 = 100$; 又 $||PF_1| - |PF_2|| = 2a = 8$. 故 $2|PF_1||PF_2| = 100 - 64 = 36$, $|PF_1||PF_2| = 18$, 面积 18.

38.[答案] $3+2\sqrt{3}$

[详解] 左焦点 $(-\sqrt{a^2+1}, 0) = (-2, 0)$, 得 $a^2 = 3$. 设 $P(x_0, y_0)$ (右支 $x_0 \geq \sqrt{3}$), $y_0^2 = x_0^2/3 - 1$. $\overrightarrow{OP} \cdot \overrightarrow{FP} = (x_0, y_0) \cdot (x_0 + 2, y_0) = x_0^2 + 2x_0 + y_0^2 = (4/3)x_0^2 + 2x_0 - 1$. 在 $[\sqrt{3}, +\infty)$ 上单调递增, $x_0 = \sqrt{3}$ 时取最小: $(4/3) \times 3 + 2\sqrt{3} - 1 = 3 + 2\sqrt{3}$.

39.[答案] (1) $x^2/400 - y^2/500 = 1$ ($x < 0$); (2) $20\sqrt{2}$ km.

[详解] (1) 以 O 为原点、 AB 所在直线为 x 轴建系: $A(-30,0)$, $B(30,0)$, $C(0,30)$. 设观察员位置 $P(x,y)$: $|PB| - |PA| = (40/V_0) \cdot V_0 = 40 < |AB| = 60$, 轨迹为双曲线左支. $2a = 40$, $2c = 60$, $a = 20$, $c = 30$, $b^2 = c^2 - a^2 = 500$, 轨迹方程 $x^2/400 - y^2/500 = 1$ ($x < 0$).

(2) 设轨迹上一点 $M(x,y)$: $|MC|^2 = x^2 + (y-30)^2 = x^2 + y^2 - 60y + 900$. 由 $x^2 = (4/5)y^2 + 400$: $|MC|^2 = (9/5)y^2 - 60y + 1300 = (9/5)(y-50/3)^2 + 800 \geq 800$. 当 $y = 50/3$ 时 $|MC|_{\min} = 20\sqrt{2}$, 扫描半径 r 至少 $20\sqrt{2}$ km.

40.[答案] CD

[详解] A:按定义实虚轴互换,共轭双曲线应为 $y^2/b^2 - x^2/a^2 = 1$, A 错; B:原曲线渐近线 $y = \pm(b/a)x$,共轭曲线(y 轴型)渐近线 $y = \pm(b/a)x$ (a/b 与 b/a 互换后方程相同),渐近线相同, B 错; C: $e_1 = c/a, e_2 = c/b, e_1 e_2 = c^2/(ab) = (a^2 + b^2)/(ab) = a/b + b/a \geq 2$ (当且仅当 $a = b$ 取等), C 对; D:焦点 $(\pm c, 0)$ 与 $(0, \pm c)$ 均在圆 $x^2 + y^2 = c^2$ 上, D 对. 故选 CD.

41.[答案] D

[详解] 设 $P(x_0, y_0), Q(-x_0, -y_0)$, $\triangle APQ$ 三顶点纵坐标和为 $y_0 - y_0 + 0 = 0$, 坐标和为 $(a, 0)$, 其重心为 $G(a/3, 0)$. 重心在每条中线上, 故 G 在中线 QE 上; QE 与 x 轴的交点唯一, 故 $M(1, 0)$ 即重心: $a/3 = 1, a = 3. b^2 = 1, c = \sqrt{9+1} = \sqrt{10}, e = c/a = \sqrt{10}/3$, 故选 D.

42.[答案] $(0, 3\sqrt{3})$

[详解] $e = 2, c^2 = 4a^2, b^2 = c^2 - a^2 = 3a^2, b/a = \sqrt{3}. k_1 k_2 = [y_0/(x_0 + a)] \cdot [y_0/(x_0 - a)] = y_0^2/(x_0^2 - a^2) = (b^2/a^2)(x_0^2 - a^2)/(x_0^2 - a^2) = b^2/a^2 = 3$ (坐标点差法, 同例 24). P 在第一象限右支: $k_3 = y_0/x_0 \in (0, b/a) = (0, \sqrt{3})$ (P 趋近顶点时 $k_3 \rightarrow 0$, 沿渐近线方向 $k_3 \rightarrow \sqrt{3}$). 故 $m = 3k_3 \in (0, 3\sqrt{3})$.

43.[答案] $[\sqrt{3}, 2\sqrt{3})$

[详解] 机器人到 P 的距离比到 Q 大 2: 距离差 $2 < |PQ| = 4$, 且距 P 更远, 轨迹为双曲线右支: $a = 1, c = 2, b^2 = 3$, 即 $x^2 - y^2/3 = 1 (x \geq 1)$. 直线 $y - 3 = k(x - \sqrt{3}), k = \sqrt{3}$ 时直线恰为渐近线 $y = \sqrt{3}x$, 符合“接触不到”; $3 - k^2 \neq 0$ 时由 $\Delta < 0$ 解得 $\sqrt{3} < k < 2\sqrt{3} (k = -\sqrt{3}$ 不合). 综上 $k \in [\sqrt{3}, 2\sqrt{3})$. 验算: $k = 2\sqrt{3}$ 时 $\Delta = 0$, 直线与右支相切(可触), 故右端开合.

44.[答案] A

[详解] $e = 2, b^2/a^2 = 3$, 直线 $MN: y = \sqrt{3}(x + a)$. 设 $P(t, \sqrt{3}t + \sqrt{3}a), t \in [-a, 0], \overrightarrow{PF_1} \cdot \overrightarrow{PF_2} = (-c - t, -\sqrt{3}t - \sqrt{3}a) \cdot (c - t, -\sqrt{3}t - \sqrt{3}a) = t^2 - c^2 + 3(t + a)^2 = 4t^2 + 6at - a^2 = 4(t + 3a/4)^2 - 13a^2/4$. $t = -3a/4$ 取最小($y_P = \sqrt{3}a/4$), $t = 0$ 取最大($y_P = \sqrt{3}a$). 面积比即纵坐标绝对值之比: $S_1/S_2 = \sqrt{3}a/(\sqrt{3}a/4) = 4$, 故选 A.

45.[答案] 2

[详解] $D(a, b), E(a, -b), |DE| = 2b, O$ 到直线 $x = a$ 距离为 $a: S = (1/2) \cdot a \cdot 2b = ab$. 焦距 4, $c = 2, a^2 + b^2 = 4 \geq 2ab$, 得 $ab \leq 2 (a = b = \sqrt{2}$ 取等). $\triangle ODE$ 面积最大值 2.

46.[答案] $\sqrt{5}$

[详解] 以最细处直径 A_1A_2 为 x 轴、中垂线为 y 轴建系: $2a = 16, a = 8$. 上瓶口点 $B(10, 12)$ 在双曲线 $x^2/64 - y^2/b^2 = 1$ 上: $100/64 - 144/b^2 = 1, 144/b^2 = 36/64, b^2 = 256. c^2 = a^2 + b^2 = 64 + 256 = 320, e^2 = 320/64 = 5, e = \sqrt{5}$.

47.[答案] 在 A 的北偏东 30° 方向, 相距 10 海里处。

[详解] 以 AB 为 x 轴、 AB 中点为原点建系: $A(3, 0), B(-3, 0), C(-5, 2\sqrt{3})$. $|PB| - |PA| = 4 < 6 = |AB|, P$ 在以 A, B 为焦点、实轴长 4 的双曲线右支上: $x^2/4 - y^2/5 = 1 (x \geq 2)$.

$|PB| = |PC|$ (B, C 同时收到), P 在 BC 中垂线上: BC 中点 $(-4, \sqrt{3}), k_{BC} = 2\sqrt{3}/(-5+3) = -\sqrt{3}$, 中垂线 $y - \sqrt{3} = (\sqrt{3}/3)(x + 4)$, 即 $x - \sqrt{3}y + 7 = 0$. 与双曲线联立($y = (x+7)/\sqrt{3}$ 代入): $11x^2 - 56x - 256 = 0$, 解得 $x = 8$ 或 $x = -32/11$ (左支交点, 不合 $x \geq 2$, 舍), $y = 5\sqrt{3}. P(8, 5\sqrt{3}): k_{AP} = 5\sqrt{3}/(8-3) = \sqrt{3}$, 倾斜角 $60^\circ, |PA| = 10$. 故 P 在 A 的北偏东 30° 方向, 相距 10 海里处. 验算: $|PB| = \sqrt{11^2 + 7^2} = 14, |PB| - |PA| = 4$ 合.

48.[答案] 2

[详解] 两者焦点均在 x 轴: $9-k^2=k+3 (k>0)$, 即 $k^2+k-6=0, (k+3)(k-2)=0, k=2 (k=-3 \text{ 舍})$. 验算: $k=2$ 时椭圆 $c^2=9-4=5$, 双曲线 $c^2=2+3=5$ 合.

49.[答案] C

[详解] 以 O 为原点、 AD 所在直线为 x 轴建系. 八等分结合 $AB=1, BC=2, CD=1$: 圆半径 $R=2$, 双曲线过点 $(\sqrt{2}, \sqrt{2}), a=1, 2/1-2/b^2=1$, 得 $b^2=2, c^2=1+2=3, c=\sqrt{3}$, 焦距 $2\sqrt{3}$, 故选 C.

50.[答案] B

[详解] $\overrightarrow{P_0F_1} \cdot \overrightarrow{PF_1} = 0$, 即 $P_0F_1 \perp PF_1$; P_0 与 P 关于原点对称, $|P_0F_2|=|PF_1|=a$ 且 $P_0F_1 \perp P_0F_2 (P_0F_2 \parallel PF_1)$. P_0 在右支: $|P_0F_1|-|P_0F_2|=2a$, 得 $|P_0F_1|=3a$. Rt $\triangle P_0F_1F_2$ (直角在 P_0): $(3a)^2+a^2=4c^2, 10a^2=4(a^2+b^2), b^2/a^2=3/2$. 渐近线 $y=\pm(\sqrt{6}/2)x$, 故选 B.

51.[答案] $(-\infty, -2]$

[详解] 设 $P(x_0, y_0) (|x_0| \geq \sqrt{3}), Q(-x_0, -y_0)$. $\overrightarrow{MP} = (x_0, y_0-1), \overrightarrow{MQ} = (-x_0, -y_0-1), \overrightarrow{MP} \cdot \overrightarrow{MQ} = -x_0^2 - y_0^2 + 1$. 由 $y_0^2 = x_0^2/3 - 1: \overrightarrow{MP} \cdot \overrightarrow{MQ} = -x_0^2 - x_0^2/3 + 1 + 1 = 2 - (4/3)x_0^2 \leq 2 - (4/3) \times 3 = -2$. 取值范围 $(-\infty, -2]$.

52.[答案] 最大值为 $5/3$.

[详解] $|PF_1|-|PF_2|=3|PF_2|=2a$, 得 $|PF_2|=(2/3)a$. 右支焦半径 $|PF_2| \geq c-a: (2/3)a \geq c-a, c \leq (5/3)a, e \leq 5/3$; 又 $e > 1$. 故 e 的最大值 $5/3$.

53.[答案] A

[详解] $e^2 = 3/2 = c^2/a^2 = (a^2+b^2)/a^2$, 得 $b^2 = a^2/2$. 过 $(1, -2): 4/a^2 - 1/b^2 = 1, 4/a^2 - 2/a^2 = 1, a^2 = 2, b^2 = 1$. 方程 $y^2/2 - x^2 = 1$, 故选 A. 验算: $4/2 - 1/1 = 1$ 合, $e = \sqrt{3/2} = \sqrt{6}/2$ 合.

54.[答案] (1) $e=\sqrt{5}/2$; (2) 证明见解析, 定点坐标为 $(-10/3, 0)$.

[详解] (1) $a=2, b=1, c=\sqrt{4+1}=\sqrt{5}, e=c/a=\sqrt{5}/2$.

(2) 设 $A(x_1, y_1), B(x_2, y_2)$. 联立 $y=kx+m$ 与 $x^2/4 - y^2=1: (1-4k^2)x^2 - 8mkx - 4(m^2+1)=0$, 其中 $1-4k^2 \neq 0$ (否则 l 与渐近线平行, 不合两交点), $\Delta = 64m^2k^2 + 16(1-4k^2)(m^2+1) > 0; x_1+x_2 = 8mk/(1-4k^2), x_1x_2 = -4(m^2+1)/(1-4k^2); y_1y_2 = (kx_1+m)(kx_2+m) = k^2x_1x_2 + mk(x_1+x_2) + m^2 = (m^2-4k^2)/(1-4k^2)$.

以 AB 为直径的圆过 $D(-2, 0) \Leftrightarrow \overrightarrow{DA} \cdot \overrightarrow{DB} = 0 \Leftrightarrow (x_1+2)(x_2+2) + y_1y_2 = 0$, 即 $x_1x_2 + 2(x_1+x_2) + 4 + y_1y_2 = 0$. 通分 (分母 $1-4k^2$): $[m^2-4k^2] + [-4(m^2+1)] + [16mk] + [4(1-4k^2)] = -3m^2 + 16mk - 20k^2 = 0$, 即 $3m^2 - 16mk + 20k^2 = 0$, 解得 $m=2k$ 或 $m=(10/3)k$. 当 $m=2k$ 时, $l: y=k(x+2)$ 过 $D(-2, 0)$, 此时 A, B 之一即 D , 与“ A, B 均异于左、右顶点”矛盾, 舍去; 当 $m=(10/3)k$ 时 ($k \neq 0$), $l: y=k(x+10/3)$ 恒过定点 $(-10/3, 0)$, 且不过 D , 符合题意. 故直线 l 过定点, 定点坐标为 $(-10/3, 0)$. 验算: $3m^2 - 16mk + 20k^2 = (3m-10k)(m-2k)$ 合; 定点 $(-10/3, 0)$ 在左顶点 $D(-2, 0)$ 左侧, 直线族绕该点转动时与双曲线恒有两交点 (k 充分小时), 满足 $\Delta > 0$ 合.

55.[答案] (1) $y=\pm\sqrt{3}x$; (2) 存在, 定直线方程 $x=1/2$.

[详解] (1) $|OF_1|=|OF_2|=c=2, |OP|=|OF_1|$ 说明 $\triangle PF_1F_2$ 中边 F_1F_2 上的中线等于该边一半, 故 $\angle F_1PF_2=90^\circ$, 即 $PF_1 \perp PF_2$. P 在右支: $|PF_1|-|PF_2|=2a$; 面积: $|PF_1| \cdot |PF_2|=2S=6. |PF_1|^2 + |PF_2|^2 = |F_1F_2|^2 = (2c)^2 = 16 = (|PF_1|-|PF_2|)^2 + 2|PF_1||PF_2| = 4a^2 + 12$, 得 $4a^2=4, a=1. b^2=c^2-a^2=4-1=3$, 渐近线 $y=\pm(b/a)x=\pm\sqrt{3}x$.

(2) 双曲线 $C: x^2 - y^2/3 = 1$. (i) l 斜率不存在: $l: x = 2$, $M(2, 3), N(2, -3)$. 直线 $A_1M: y = x + 1$; 直线 $A_2N: y = -3x + 3$. 联立得 $Q(1/2, 3/2)$, 横坐标 $1/2$.

(ii) l 斜率存在: l 过 $F_2(2, 0)$ 且不与渐近线平行, 设 $l: y = k(x - 2) (k \neq 0, k \neq \pm\sqrt{3})$, $M(x_1, y_1), N(x_2, y_2)$. 联立得 $(3 - k^2)x^2 + 4k^2x - 4k^2 - 3 = 0$, $x_1 + x_2 = 4k^2/(k^2 - 3)$, $x_1x_2 = (4k^2 + 3)/(k^2 - 3)$. 直线 $A_1M: y = [y_1/(x_1 + 1)](x + 1)$; 直线 $A_2N: y = [y_2/(x_2 - 1)](x - 1)$. 交点 $Q(x, y)$ 满足 $(x + 1)/(x - 1) = y_2(x_1 + 1)/[y_1(x_2 - 1)] (x \neq 1)$. 两边平方, 以 $y_1^2 = 3(x_1^2 - 1), y_2^2 = 3(x_2^2 - 1)$ 代入: $[(x + 1)/(x - 1)]^2 = [3(x_2^2 - 1)(x_1 + 1)^2]/[3(x_1^2 - 1)(x_2 - 1)^2] = [(x_2 + 1)(x_1 + 1)]/[(x_1 - 1)(x_2 - 1)] = [x_1x_2 + (x_1 + x_2) + 1]/[x_1x_2 - (x_1 + x_2) + 1] = [(4k^2 + 3) + 4k^2 + (k^2 - 3)]/[(4k^2 + 3) - 4k^2 + (k^2 - 3)] = 9k^2/k^2 = 9$. 故 $(x + 1)/(x - 1) = \pm 3$, 得 $x = 1/2$ 或 $x = 2$.

取 $x = 2$ 需 $y_2(x_1 + 1)/[y_1(x_2 - 1)] = +3 > 0$; 但 M, N 是过右焦点的弦的两端点且均非顶点: $0 < k^2 < 3$ 时 M, N 分居两支 ($x_1x_2 < 0$), $x_2 - 1 < 0$ 而 $y_1y_2 > 0$; $k^2 > 3$ 时 M, N 同在右支且分居 x 轴两侧, $y_1y_2 < 0$ 而 $(x_1 + 1)(x_2 - 1) > 0$ ——两种情形该式恒为负, 只能取 -3 . 故 $x = 2$ 为平方增根, 舍去, $x = 1/2$. 综上, 交点 Q 恒在定直线 $x = 1/2$ 上. 验算: (i) 的 $Q(1/2, 3/2)$ 落在 $x = 1/2$ 上合; (ii) 韦达分子分母化简 $4k^2 + 3 + 4k^2 + k^2 - 3 = 9k^2, 4k^2 + 3 - 4k^2 + k^2 - 3 = k^2$, 比值 9 合.

56.[答案] C

[详解] 设 $PQ: x = my + b$, 联立 $x^2 - y^2/2 = 1: (m^2 - 1/2)y^2 + 2bmy + b^2 - 1 = 0$. 设 $P(x_1, y_1), Q(x_2, y_2)$: $y_1 + y_2 = -2bm/(m^2 - 1/2), y_1y_2 = (b^2 - 1)/(m^2 - 1/2)$. $AP \perp AQ (A(-1, 0))$: $(x_1 + 1)(x_2 + 1) + y_1y_2 = 0$, 以 $x_i = my_i + b$ 代入: $(1 + m^2)y_1y_2 + (b + 1)m(y_1 + y_2) + (b + 1)^2 = 0$.

通分 (乘 $m^2 - 1/2$): $2(b^2 - 1)(1 + m^2) - 4bm^2(b + 1) + (2m^2 - 1)(b + 1)^2 = 0$. 按 m 整理—— m^2 系数: $2(b^2 - 1) - 4b(b + 1) + 2(b + 1)^2 = 2b^2 - 2 - 4b^2 - 4b + 2b^2 + 4b + 2 = 0$ (恒为 0); 常数项: $2(b^2 - 1) - (b + 1)^2 = b^2 - 2b - 3 = 0$, 得 $b = 3$ 或 $b = -1$. $b = -1$ 时 PQ 过 $A(-1, 0) (P, Q$

之一与 A 重合或直线退化), 舍去; $b = 3, PQ$ 恒过 $(3, 0)$. 故选 C.

验算 (特例): $PQ \perp x$ 轴即 $m = 0, x = 3$: 与 $x^2 - y^2/2 = 1$ 交于 $(3, \pm 4); \overrightarrow{AP} = (4, 4), \overrightarrow{AQ} = (4, -4), \overrightarrow{AP} \cdot \overrightarrow{AQ} = 16 - 16 = 0$ 合.

57.[答案] (1) $x^2 - y^2/3 = 1$; (2) 证明见解析, $M(-1, 0)$.

[详解] (1) 取渐近线 $y = (b/a)x$, 与 $x = a^2/c$ 交于 $P(a^2/c, ab/c)$. $|PF|^2 = (c - a^2/c)^2 + (ab/c)^2$, 其中 $c - a^2/c = (c^2 - a^2)/c = b^2/c$, 故 $|PF|^2 = b^4/c^2 + a^2b^2/c^2 = b^2(a^2 + b^2)/c^2 = b^2c^2/c^2 = b^2$. 由 $|PF| = \sqrt{3}$ 得 $b^2 = 3, e^2 = c^2/a^2 = 4 = (a^2 + b^2)/a^2$, 得 $a^2 = 1, C: x^2 - y^2/3 = 1$.

(2) $F(2, 0)$. 设 $Q(x_0, y_0) (x_0 \geq 1), 3x_0^2 - y_0^2 = 3$. ① $x_0 = 2$ 时 $y_0 = \pm 3$: $\angle QFM = 90^\circ$ 需 $\angle QMF = 45^\circ, |MF| = |QF| = |y_0| = 3$, 得 $M(-1, 0)$, 符合. ② $x_0 \neq 2$ 时设 $M(t, 0) (t < 0)$. 有向正切: $\tan \angle QFM = -y_0/(x_0 - 2), \tan \angle QMF = y_0/(x_0 - t)$. $\angle QFM = 2\angle QMF$, 由倍角公式: $-y_0/(x_0 - 2) = 2[y_0/(x_0 - t)]/(1 - [y_0/(x_0 - t)]^2)$. $y_0 \neq 0$ 时约去 y_0 、交叉相乘: $-(x_0 - t)^2 - y_0^2 = 2(x_0 - 2)(x_0 - t)$, 展开整理: $3x_0^2 - y_0^2 - (4 + 4t)x_0 + 4t + t^2 = 0$. 以 $3x_0^2 - y_0^2 = 3$ 代入: $-(4 + 4t)x_0 + (t^2 + 4t + 3) = 0$, 对任意 $x_0 \geq 1 (x_0 \neq 2)$ 恒成立, 故 $-(4 + 4t) = 0$ 且 $t^2 + 4t + 3 = 0$, 得 $t = -1$ (两式公共解). $y_0 = 0 (Q$ 为右顶点) 时直接验证 $M(-1, 0)$ 亦满足 $\angle QFM = 2\angle QMF$. 综上, 存在定点 $M(-1, 0)$. 验算: $t^2 + 4t + 3 = (t + 1)(t + 3)$ 与 $-(4 + 4t)$ 公共根 $t = -1$ 合; 取 $Q(2, 3): \angle QFM = 90^\circ, \angle QMF = 45^\circ, 2 \times 45^\circ = 90^\circ$ 合.

58.[答案] (1) $x^2 - y^2/3 = 1 (x > 0)$; (2) $(-\infty, -\sqrt{3}) \cup (\sqrt{3}, +\infty)$; (3) 存在, $M(-1, 0)$.

[详解] (1) $|PF_1| - |PF_2| = 2 < |F_1F_2| = 4$, 轨迹为以 F_1, F_2 为焦点、实轴长 2 的双曲线右支: $c = 2, a = 1, b^2 = 3, \Gamma: x^2 - y^2/3 = 1 (x > 0)$.

(2) $l: y = k(x - 2)$, 代入 $\Gamma: (3 - k^2)x^2 + 4k^2x - 4k^2 - 3 = 0, A, B$ 为两交点 (右支上 $x > 0$): 需 $3 - k^2 \neq 0, \Delta > 0, x_1 + x_2 = -4k^2/(3 - k^2) > 0, x_1x_2 = (-4k^2 - 3)/(3 - k^2) > 0$.

由 $x_1 + x_2 > 0$ 得 $k^2 > 3$; 此时 $x_1 x_2 > 0$, $\Delta = 16k^4 - 4(3-k^2)(-4k^2-3) > 0$ 亦成立. 故 $k \in (-\infty, -\sqrt{3}) \cup (\sqrt{3}, +\infty)$.

(3) 存在性: 设 $M(m, 0)$. l 与 Γ 联立同上: $(3-k^2)x^2 + 4k^2x - 4k^2 - 3 = 0$ ($k^2 > 3$), $x_1 + x_2 = -4k^2/(3-k^2)$, $x_1 x_2 = (-4k^2-3)/(3-k^2)$. $MA \perp MB \Leftrightarrow (x_1-m)(x_2-m) + y_1 y_2 = 0$, $y_i = k(x_i-2): (1+k^2)x_1 x_2 - (m+2k^2)(x_1 + x_2) + m^2 + 4k^2 = 0$. 乘 $(3-k^2)$ 展开: $(1+k^2)(-4k^2-3) + 4k^2(2k^2+m) + (m^2+4k^2)(3-k^2) = 0$, 即 $k^2(-m^2+4m+5) + 3m^2-3 = 0$, 对一切 $k^2 > 3$ 恒成立. 故 $-m^2+4m+5 = 0$ 且 $3m^2-3 = 0$, 得 $m = -1$ ($m = 1$ 不满足第一式). 斜率不存在时 $l: x = 2, A(2, 3), B(2, -3); \overrightarrow{MA} = (3, 3), \overrightarrow{MB} = (3, -3), \overrightarrow{MA} \cdot \overrightarrow{MB} = 0$ 合. 综上, 存在 $M(-1, 0)$. 验算: $k^2(-m^2+4m+5) + 3m^2-3$ 在 $m = -1$ 时两项恒为 0 合; (3) 的韦达代入式与 (2) 一致合.

59. [答案] (1) $e=2$; (2) $3/2$.

[详解] (1) 右支上 $|PF_1|_{\min} = c+a$ (左顶点处), $|PF_2|_{\min} = c-a$ (右顶点处). $m_1 m_2 = c^2 - a^2 = b^2 = 3a^2$, 得 $c^2 = 4a^2, e = 2$.

(2) $a = 2, c = 4, b^2 = 12$, 双曲线 $x^2/4 - y^2/12 = 1$, $F_1(-4, 0)$. 设 $AB: y = k(x+4)$ ($k \neq \pm\sqrt{3}$). 联立: $(3-k^2)x^2 - 8k^2x - 16k^2 - 12 = 0$, $x_1 + x_2 = 8k^2/(3-k^2)$, $x_1 x_2 = (-16k^2-12)/(3-k^2)$. 弦长 $|AB| = \sqrt{1+k^2} \cdot \sqrt{(x_1+x_2)^2 - 4x_1 x_2} = 12(1+k^2)/|3-k^2|$. 中点 $(4k^2/(3-k^2), 12k/(3-k^2))$ ($y_1 + y_2 = k(x_1 + x_2) + 8k = 24k/(3-k^2)$). 垂直平分线: $y - 12k/(3-k^2) = -(1/k)(x - 4k^2/(3-k^2))$, 令 $x = 0$ 得 $y = 16k/(3-k^2)$, $|OD| = 16|k|/|3-k^2|$.

$|AB|/|OD| = [12(1+k^2)/|3-k^2|] \cdot [|3-k^2|/(16|k|)] = (3/4)(|k|+1/|k|) \geq 3/2$ ($|k| = 1$ 取等), 最小值 $3/2$. 验算: $(x_1-x_2)^2 = (8k^2/(3-k^2))^2 + 4(16k^2+12)/(3-k^2) = [64k^4+4(16k^2+12)(3-k^2)]/(3-k^2)^2 = 144(k^2+1)/(3-k^2)^2$, $|AB| = \sqrt{1+k^2} \cdot 12\sqrt{k^2+1}/|3-k^2|$ 合.

60. [答案] (1) $x^2/2 - y^2 = 1$; (2) 过定点 $(0, 1)$.

[详解] (1) $e^2 = 3/2 = c^2/a^2 = (a^2+b^2)/a^2$, 得 $b^2/a^2 = 1/2$, 即 $a^2 = 2b^2$, 双曲线方程为 $x^2/(2b^2) - y^2/b^2 = 1$. 代入 P : $3/(2b^2) - (1/2)/b^2 = 1$, $(3-1)/(2b^2) = 1$, $b^2 = 1, a^2 = 2$. $C: x^2/2 - y^2 = 1$.

(2) AB 斜率不存在时: $A(x_0, y_0), B(x_0, -y_0), k_1+k_2 = (y_0-1)/(x_0-2) + (-y_0-1)/(x_0-2) = -2/(x_0-2) = 1$, 得 $x_0 = 0$, 不在曲线上, 舍; 故斜率存在. 设 $AB: y = kx + t$, 代入: $(2k^2-1)x^2 + 4ktx + 2t^2 + 2 = 0$ ($2k^2-1 \neq 0$), $x_1 + x_2 = -4kt/(2k^2-1), x_1 x_2 = (2t^2+2)/(2k^2-1)$. $k_1+k_2 = 1: (kx_1+t-1)/(x_1-2) + (kx_2+t-1)/(x_2-2) = 1$, 即 $(2k-1)x_1 x_2 + (t-2k+1)(x_1+x_2) - 4t = 0$. 代入: $(2k-1)(2t^2+2) - 4kt(t-2k+1) - 4t(2k^2-1) = 0$ (通分后乘 $2k^2-1$), 整理 $t^2 + (2k-2)t - 1 + 2k = 0$, 即 $(t-1)(t+2k-1) = 0$. $t = 1: y = kx + 1$ 过 $(0, 1)$; $t = 1-2k: y = k(x-2) + 1$ 过 $Q(2, 1)$, 不合 (A, B 异于 Q 且 Q 为弦端情形). 故直线 AB 过定点 $(0, 1)$. 验算: $t^2 + (2k-2)t + 2k - 1 = (t-1)(t+2k-1)$ 展开核对合.

61. [答案] (1) $x^2/4 - y^2 = 1$; (2) 存在, $Q(23/8, 0), QM \cdot QN = 273/64$.

[详解] (1) $b/a = 1/2$ 且 $16/a^2 - 3/b^2 = 1: b^2 = a^2/4, 16/a^2 - 12/a^2 = 1, a^2 = 4, b^2 = 1$. $C: x^2/4 - y^2 = 1$. (2) 设 $l: x = my + 1$, 联立: $(m^2-4)y^2 + 2my - 3 = 0$ ($m^2 \neq 4, \Delta > 0$ 得 $m^2 > 3$), $y_1 + y_2 = -2m/(m^2-4), y_1 y_2 = -3/(m^2-4); x_1 + x_2 = m(y_1 + y_2) + 2 = -8/(m^2-4), x_1 x_2 = (my_1 + 1)(my_2 + 1) = -(4m^2+4)/(m^2-4)$. 设 $Q(t, 0): \overrightarrow{QM} \cdot \overrightarrow{QN} = (x_1-t)(x_2-t) + y_1 y_2 = x_1 x_2 - t(x_1 + x_2) + t^2 + y_1 y_2 = -4 + t^2 + (8t-23)/(m^2-4)$. 与 m 无关 $\Leftrightarrow 8t-23 = 0$, 得 $t = 23/8$, 常数 $-4 + t^2 = -4 + 529/64 = 273/64$. 验算: $x_1 x_2 = (-4m^2-4)/(m^2-4) = -4 - 20/(m^2-4)$; 三项合并分子 $-20 - 3 + 8t = 8t - 23$, 故数量积 $-4 + t^2 + (8t-23)/(m^2-4)$ 合.

62. [答案] (1) $15/4$; (2) 双曲线中 $k_A B \cdot k_O M = b^2/a^2$

[详解] (1) 联立得 $16x^2+12x-9=0, x_1+x_2=-3/4, x_1x_2=-9/16$. 根差 $(x_1-x_2)^2=(-3/4)^2-4\times(-9/16)=45/16$, 故 $|AB|=\sqrt{5}\cdot\sqrt{5}/4=15/4$.

(2) 性质: 双曲线 $x^2/a^2-y^2/b^2=1$ 上任意两点 A, B (斜率存在时), M 为 AB 中点, 则 $k_{AB}\cdot k_{OM}=b^2/a^2$. 证明 (点差法): $A(x_3, y_3), B(x_4, y_4), M(x_0, y_0)$. 两式相减: $(x_3+x_4)(x_3-x_4)/a^2=(y_3+y_4)(y_3-y_4)/b^2$, 即 $2x_0(x_3-x_4)/a^2=2y_0(y_3-y_4)/b^2$, 变形得 $k_{AB}\cdot k_{OM}=[(y_3-y_4)/(x_3-x_4)]\cdot(y_0/x_0)=b^2/a^2$. 定值证毕. 验算: $k_{AB}\cdot k_{OM}=b^2/a^2$ 与例 24 的 $k_1k_2=b^2/a^2$ 相容 (中心对称两点为其特例) 合.

63.[答案] (1) $x^2-y^2/8=1$; (2) 定值 $2\sqrt{2}$.

[详解] (1) $c=3, b/a=2\sqrt{2}, a^2+b^2=9$, 得 $a^2(1+8)=9$, $a^2=1, b^2=8$. $C: x^2-y^2/8=1$.

(2) $l: y=kx+m (k\neq\pm 2\sqrt{2}, m\neq 0, \text{斜率存在且不为 } 0)$. $D(-m/k, 0), |OD|=|m/k|$. 相切: 联立 $(8-k^2)x^2-2kmx-m^2-8=0, \Delta=0$, 得 $4k^2m^2+4(8-k^2)(m^2+8)=0$, 即 $m^2=k^2-8 (k^2>8)$. M 与 $y=2\sqrt{2}x$ 交: $(m/(2\sqrt{2}-k), 2\sqrt{2}m/(2\sqrt{2}-k))$; N 与 $y=-2\sqrt{2}x$ 交: $(-m/(2\sqrt{2}+k), 2\sqrt{2}m/(2\sqrt{2}+k))$. $S_{\triangle MON}=S_{\triangle MOD}+S_{\triangle NOD}=(1/2)|OD|\cdot|y_M-y_N|=(1/2)|m|\cdot|x_M-x_N|=(1/2)|m|\cdot|4\sqrt{2}m/(8-k^2)|=2\sqrt{2}m^2/(k^2-8)=2\sqrt{2}$ (定值). 验算: $m^2=k^2-8>0$ 与 $k^2>8$ 一致合; S 只依赖 $m^2/(k^2-8)$, 化简后为常数合.

64.[答案] (1) $y=\pm\sqrt{3}x$; (2) 定值 $k_1/k_2=-1/3$.

[详解] (1) $a=1, c=2, b^2=3, \Gamma: x^2-y^2/3=1$, 渐近线 $y=\pm\sqrt{3}x$.

(2) l 斜率不存在: $P(2, 3), Q(2, -3), k_1=3/3=1, k_2=-3/1=-3$, 比值 $-1/3$. $l: y=k(x-2) (k\neq 0)$: $(k^2-3)x^2-4k^2x+4k^2+3=0, x_1+x_2=4k^2/(k^2-3), x_1x_2=(4k^2+3)/(k^2-3)$. $3k_1+k_2=3y_1/(x_1+1)+y_2/(x_2-1)=k[4x_1x_2-5(x_1+x_2)+4]/[(x_1+1)(x_2-1)]$. $4x_1x_2-5(x_1+x_2)+4=[4(4k^2+3)-20k^2+4(k^2-3)]/(k^2-3)=0$. 故 $3k_1+k_2=0, k_1/k_2=-1/3$ (定值), 综上得证. 验算: 分子 $16k^2+12-20k^2+4k^2-12=0$ 合.

65.[答案] (1) $2x^2-y^2/2=1$; (2) 定直线 $12x-y-2=0$.

[详解] (1) $2b=2\sqrt{2}$, 得 $b^2=2; e^2=5=1+b^2/a^2$, 得 $a^2=1/2$. $C: 2x^2-y^2/2=1$.

(2) 设 $M(x, y), A(x_1, y_1), B(x_2, y_2) (x_1<x_2<3)$. $|AM|/|MB|=|AP|/|PB|$ 结合共线定比: $(3-x_1)/(x_2-3)=-(x-x_1)/(x_2-x)$, 即 $[6-(x_1+x_2)]x=3(x_1+x_2)-2x_1x_2$. ① 设 $l: y-1=k(x-3)$, 代入 $C: (4-k^2)x^2+(6k^2-2k)x-9k^2+6k-3=0, x_1+x_2=(2k-6k^2)/(4-k^2), x_1x_2=(-9k^2+6k-3)/(4-k^2)$. 代入①整理: $12x-3=k(x-3)$, 与 $y-1=k(x-3)$ 联立消 k : $12x-3=y-1$, 即 $12x-y-2=0$. 点 M 恒落在定直线 $12x-y-2=0$ 上. 验算: 由定比 $x=[3(x_1+x_2)-2x_1x_2]/[6-(x_1+x_2)]$ (分比不变) 合; 韦达代入化简得 $12x-3=k(x-3)$, 代回 $y-1=k(x-3)$ 消 k 得 $y=12x-2$ 合.

66.[答案] (1) $2\sqrt{5}x-y-4\sqrt{5}=0$ 或 $2\sqrt{5}x+y-4\sqrt{5}=0$; (2) 定直线 $x=1/2$.

[详解] 设 $l: x=my+2, C(x_1, y_1), D(x_2, y_2)$, 联立: $(4m^2-1)y^2+16my+12=0, y_1+y_2=-16m/(4m^2-1), y_1y_2=12/(4m^2-1)$.

(1) $|CN|=3|ND| (N \text{ 为 } F_2, C, D \text{ 同向分段})$, 得 $y_1=-3y_2, y_2=8m/(4m^2-1)$; 代入 $y_1y_2=-3y_2^2=12/(4m^2-1): -3\times 64m^2/(4m^2-1)^2=12/(4m^2-1)$, 得 $-16m^2=4m^2-1, m^2=1/20, m=\pm\sqrt{5}/10$. 直线 $l: x=\pm(\sqrt{5}/10)y+2$, 即 $2\sqrt{5}x-y-4\sqrt{5}=0$ 或 $2\sqrt{5}x+y-4\sqrt{5}=0$.

(2) $AC: y=[y_1/(x_1+1)](x+1); BD: y=[y_2/(x_2-1)](x-1)$. 以 $x_1=my_1+2, x_2=my_2+2$ 代入, 得 $x_1+1=my_1+3, x_2-1=my_2+1$. 联立: $(x+1)/(x-1)=(my_1y_2+3y_2)/(my_1y_2+y_1)$. 由韦达 $y_1y_2=12/(4m^2-1), y_1+y_2=-16m/(4m^2-1)$, 得 $y_1y_2=-[3/(4m)](y_1+y_2)$. 于是 $(x+1)/(x-1)=[-(3/4)(y_1+y_2)+3y_2]/[-(3/4)(y_1+y_2)+y_1]=(-3y_1+9y_2)/(y_1-3y_2)=-3$, 解得 $x=1/2$. 点 P 恒在定直线 $x=1/2$ 上. 验算: $(-3y_1+9y_2)/(y_1-3y_2)=-3(y_1-3y_2)/(y_1-3y_2)=-3$ 合; (1) 中 $-16m^2=4m^2-1$ 即 $20m^2=1$ 合.

67.[答案] A

[详解] $e^2 = 1 + b^2/a^2 = 1 + m/9 = 13/9$, 得 $m/9 = 4/9$, $m = 4$. 故 $C: x^2/9 - y^2/4 = 1$, $a = 3, b = 2$, 渐近线 $y = \pm(b/a)x = \pm(2/3)x$, 选 A. 验算: $e = \sqrt{1+4/9} = \sqrt{13}/3$ 合.

68.[答案] A

[详解] 逐项核两条件: A: $a^2 = 4, b^2 = 2$, 渐近线 $y = \pm(b/a)x = \pm(\sqrt{2}/2)x$, $c^2 = 4 + 2 = 6$, 焦点 $(\pm\sqrt{6}, 0)$ 在 x 轴, 两条件皆合; B: 渐近线 $y = \pm(a/b)x = \pm(\sqrt{2}/2)x$ 与 A 相同, 但实轴在 y 轴上, 焦点 $(0, \pm\sqrt{6})$, 不合“焦点在 x 轴”; C: 渐近线 $y = \pm(b/a)x = \pm\sqrt{2}x$, 不合; D: 渐近线 $y = \pm(a/b)x = \pm(1/2)x$ 且焦点 $(0, \pm2\sqrt{5})$ 在 y 轴, 两条件皆不合. 故选 A.

69.[答案] A

[详解] 由 $e^2 = 1 + (b/a)^2$ 得 $(b/a)^2 = e^2 - 1$. $e \in [\sqrt{3}, 2]$, 则 $e^2 \in [3, 4]$, $(b/a)^2 \in [2, 3]$; 又 $b/a > 0$, 故 $b/a \in [\sqrt{2}, \sqrt{3}]$, 选 A. 验算: $e = \sqrt{3}$ 时 $b/a = \sqrt{2}$, $e = 2$ 时 $b/a = \sqrt{3}$, 端点可取合.

70.[答案] $x^2/16 - y^2/4 = 1$

[详解] 虚轴长 $2b = 4$, 得 $b = 2$; 焦点在 x 轴时渐近线为 $y = \pm(b/a)x$, 由 $b/a = 1/2$ 得 $a = 2b = 4$. 故标准方程为 $x^2/16 - y^2/4 = 1$. 验算: $a^2 = 16, b^2 = 4$, 渐近线 $y = \pm(2/4)x = \pm(1/2)x$ 合, 虚轴 $2b = 4$ 合, $c^2 = 20$, 焦点 $(\pm2\sqrt{5}, 0)$ 在 x 轴合.

71.[答案] $m=14; e=2\sqrt{2}$

[详解] C_2 ($a^2 = 1, b^2 = 7$) 的渐近线为 $y = \pm\sqrt{7}x$. C_1 ($a^2 = 2, b^2 = m$, 焦点同在 x 轴) 的渐近线为 $y = \pm(\sqrt{m}/\sqrt{2})x$. 渐近线相同即 $\sqrt{m}/\sqrt{2} = \sqrt{7}$, 得 $\sqrt{m} = \sqrt{14}$, $m = 14$. 此时 $e^2 = 1 + b^2/a^2 = 1 + 14/2 = 8$, 故 $e = 2\sqrt{2}$. 验算: $m = 14$ 时 C_1 渐近线 $y = \pm(\sqrt{14}/\sqrt{2})x = \pm\sqrt{7}x$ 合, $e = \sqrt{8} = 2\sqrt{2}$ 合.

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